

Machine Learning Course GUI Report

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****Course:**** MKT3434 – Mechatronic Systems Design

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1. Overview

This GUI application was developed as part of the MKT3434 Machine Learning course. It enables experimentation with classical and modern machine learning techniques, visualization of results, and evaluation of the impact of preprocessing methods such as missing value handling and scaling.

2. Comparison of Missing Value Handling Methods

Various strategies for handling missing data were compared using the `SimpleImputer` from scikit-learn. For regression tasks (e.g., Diabetes dataset), the impact of each imputation strategy was measured using MSE.

Below is a sample result from Diabetes Dataset:(Example)

- Mean Imputation : MSE = 2808.6226
- Median Imputation : MSE = 2810.0399
- Most Frequent Imputation: MSE = 2799.4935
- Constant Fill : MSE = 2805.8840

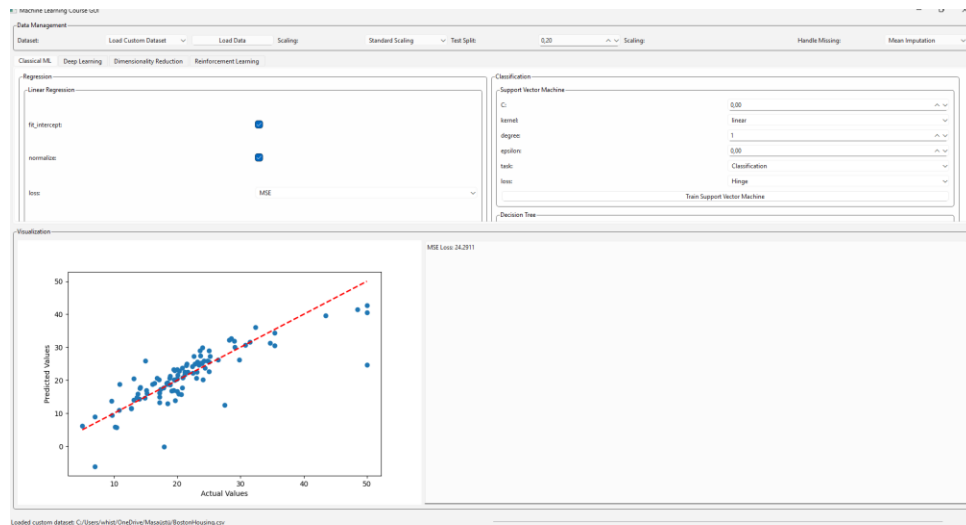
3. GUI Screenshots and Descriptions

BostonHousing.csv – Linear Regression with Different Losses

MSE (Mean Squared Error)

Below is a sample result:

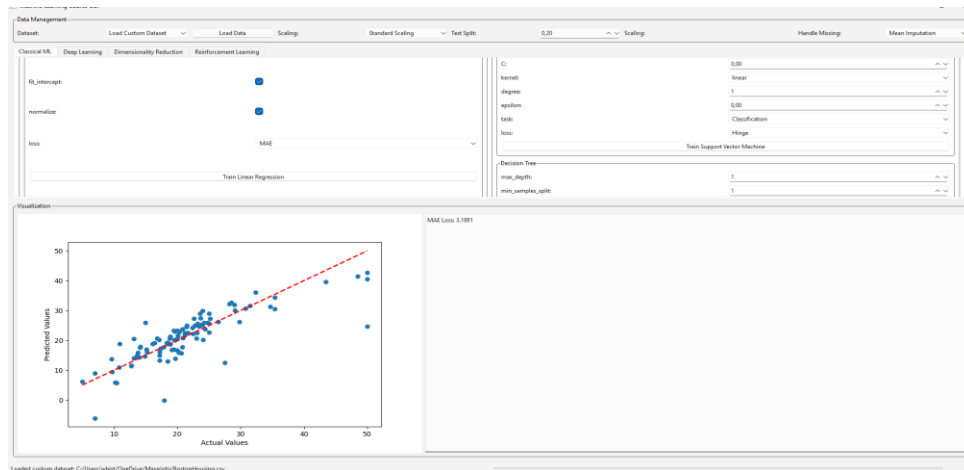
- **Dataset:** BostonHousing (custom CSV)
- **Model:** Linear Regression
- **Loss Function:** MSE
- **Result:**
MSE Loss: 24.2911



MAE (Mean Absolute Error)

Below is a sample result:

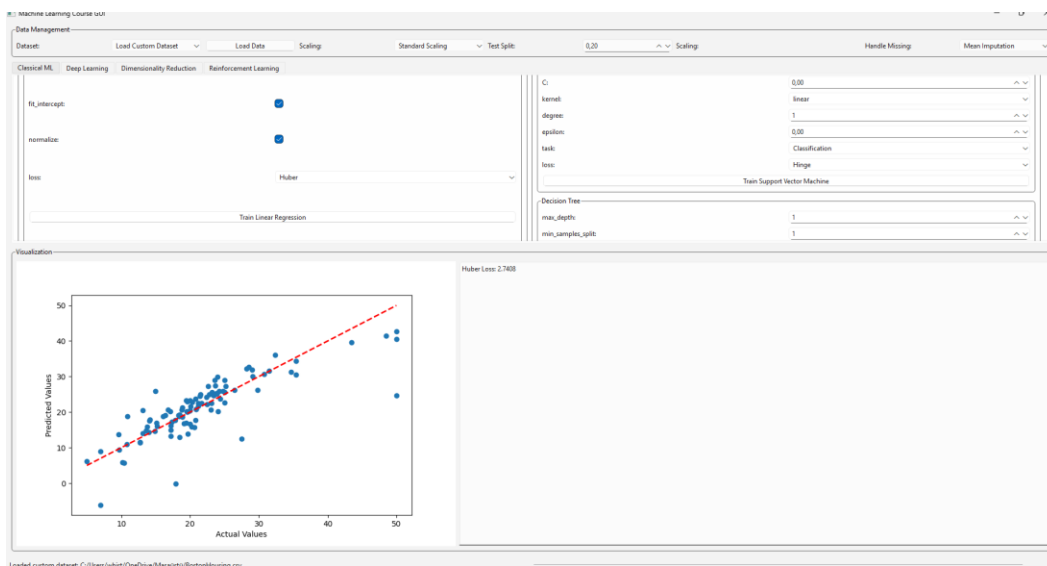
- **Dataset:** BostonHousing (custom CSV)
- **Model:** Linear Regression
- **Loss Function:** MAE
- **Result:**
MAE Loss: 3.1891



Huber Loss

Below is a sample result:

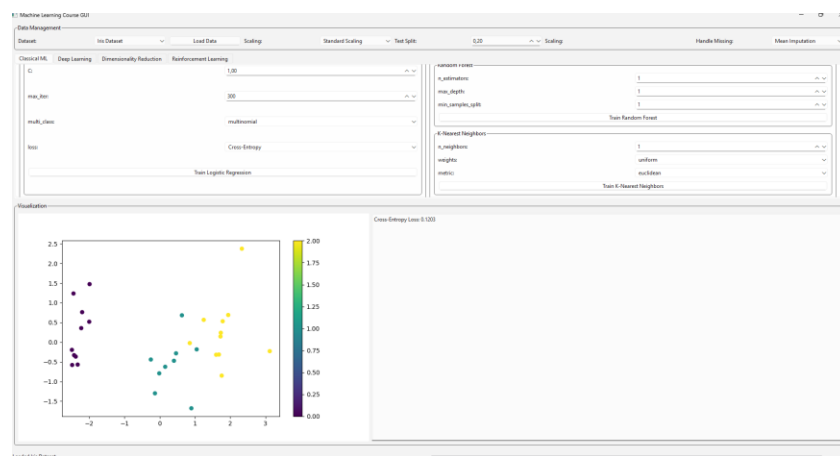
- **Dataset:** BostonHousing (custom CSV)
- **Model:** Linear Regression
- **Loss Function:** Huber Loss
- **Result:**
Huber Loss: 2.7408



3. Iris Dataset – Logistic Regression with Cross-Entropy Loss

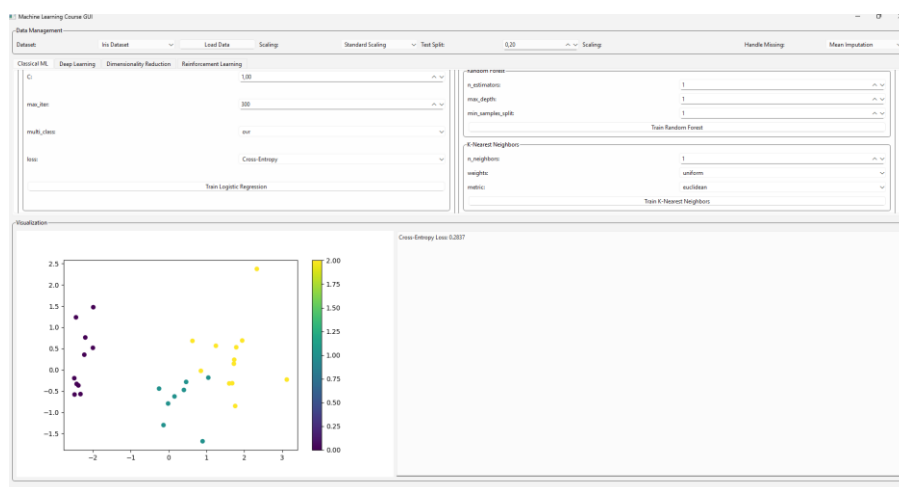
Below is a sample result:

- **Dataset:** Iris
- **Model:** Logistic Regression
- **Multi-Class Mode:** multinomial
- **Loss Function:** Cross-Entropy
- **Result:**
Cross-Entropy Loss: 0.1203



(With One-vs-Rest)

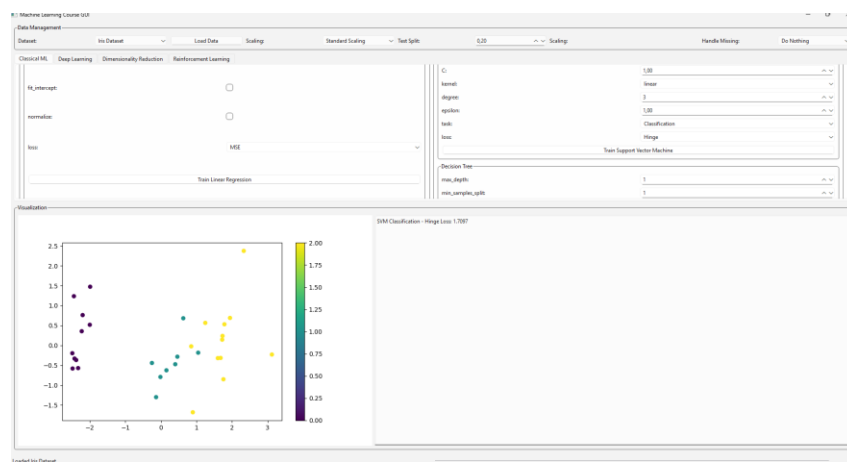
- **Loss:**
Cross-Entropy Loss: 0.2837



4. Iris Dataset – SVM with Hinge Loss

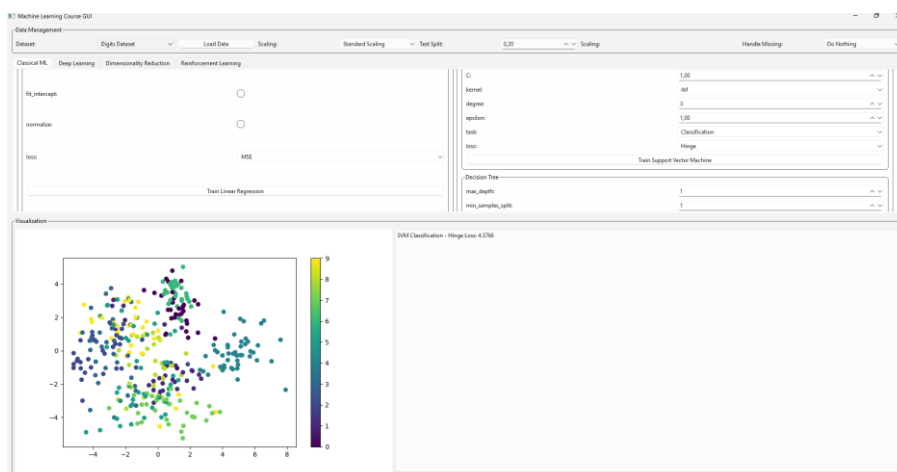
Below is a sample result:

- **Dataset:** Iris
- **Model:** SVM
- **Kernel:** Linear
- **Loss Function:** Hinge
- **Result:**
Hinge Loss: 1.7097



(With epsilon = 0.1 and degree = 1)

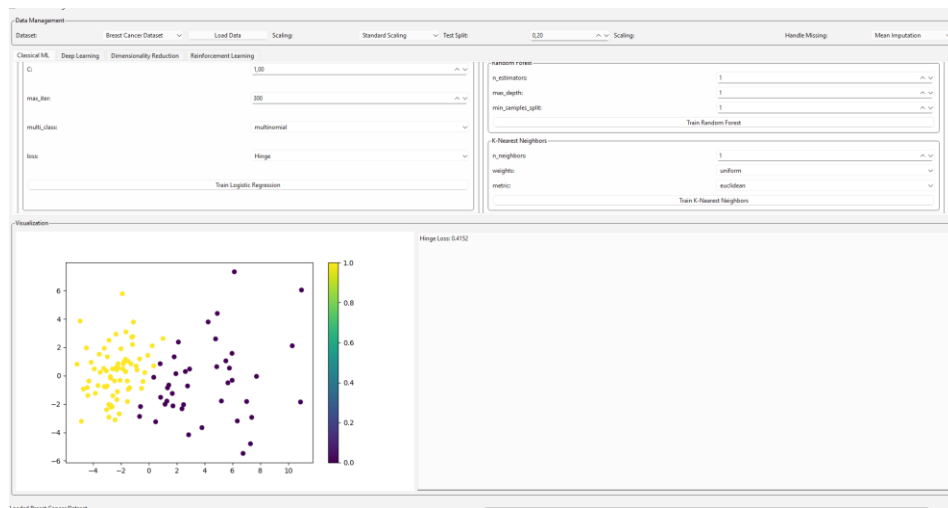
- **Result:**
Hinge Loss: 1.7443



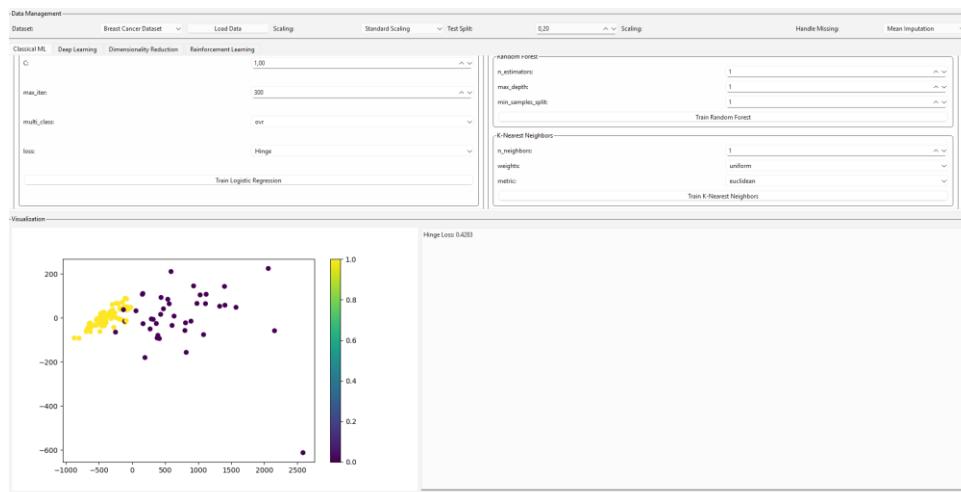
5. Breast Cancer Dataset – Logistic Regression with Hinge Loss

Below is a sample result:

- **Dataset:** Breast Cancer
- **Model:** Logistic Regression
- **Loss Function:** Hinge
- **Result (Multinomial):**
Hinge Loss: 0.4152



- **Result (OvR):**
Hinge Loss: 0.4283



Comparison of Missing Data Handling Methods

Dataset: Diabetes

Strategy	MSE
Mean	2808.6226
Median	2810.0399
Most Frequent	2799.4935
Constant (0)	2805.8840

Dataset: BostonHousing

Strategy	MSE
Mean	24.2911
Median	24.4812
Most Frequent	25.0043
Constant (0)	25.7398

Dataset: Breast Cancer

Not suitable for Linear Regression MSE comparison due to classification task and categorical target values.

Dataset: Digits

Same as above — not suitable for regression (target is categorical 0–9). MSE comparison not meaningful.

Dataset: Iris

Similar to Digits, not suited for regression-based imputation analysis.

Conclusion

- **Best Dataset for Missing Value Strategy Comparison:**
Diabetes and **BostonHousing** (continuous regression targets)
- **Best Performing Strategy on Diabetes:**
Most Frequent (lowest MSE)
- **Best Performing Strategy on BostonHousing:**
Mean (lowest MSE)